

ROUTE RESILIENCE

Occlusion-Robust Road Extraction & Graph-Theoretic Criticality Analysis

Build Roadmap, Dataset Verification & 4-Person Work Split | Hack2Skill — Bharatiya Antariksh Hackathon (BAH) 2026, Problem Statement 4

Prepared for: 4-member team | Target: Working prototype + Idea PPT, submission by **1 July** | Verified: 24 June 2026

1. Executive Summary & Strategy

The problem statement has four cleanly separable layers: (1) a Deep Learning segmentation model that recovers roads under occlusion, (2) a graph-healing step that converts broken pixel masks into a routable vector network, (3) network science (centrality + node ablation) to find weak points, and (4) a dashboard that makes all of this clickable for a planner. This maps almost perfectly onto a 4-person team — each person owns one layer end-to-end, with two integration checkpoints where the layers get wired together.

The biggest real risk is not the algorithms — it's data access. Cartosat-3 (the headline dataset in the problem statement) is **only handed out during the live 30-hour hackathon** and is not available to you today. Section 3 gives the verified workaround so this does not block the prototype.

2. Dataset Access — Verified 24 June 2026

Each dataset below was checked directly (official pages / docs) before including it in this plan.

Dataset	Status	How to actually get it	Caveat
SpaceNet Roads (SN3 & SN5)	■ Free, open	Hosted as an AWS Public Dataset. Needs a free AWS account + AWS CLI: <code>aws s3 cp s3://spacenet-dataset/spacenet/SN3_roads/...</code>	Sub-meter res. (~0.3–0.5m) — closest open proxy to Cartosat-3 resolution. Some legacy buckets are 'Requester Pays' but SN3/SN5 road tarballs download free.
DeepGlobe Road Extraction	■ Free, open	Kaggle dataset 'deepglobe-road-extraction-dataset' — just needs a Kaggle login, instant download (~3.7 GB).	6,226 images, 50cm/px DigitalGlobe imagery. Great for occlusion-robustness training.
OpenSatMap	■ Free, open	Hugging Face (z-hb/OpenSatMap) or Baidu Drive.	0.15–0.3m/px, but covers only 8 Chinese cities. Use for pretraining/augmentation only, not India validation.
OpenStreetMap road vectors	■ Free, no login	OSMnx / Overpass API — pull Bengaluru road graph directly in Python.	Use this as your real India ground truth for the graph/criticality phase.
Sentinel-2 (10m)	■ Free, open	Copernicus Data Space Ecosystem or Google Earth Engine — no ISRO account needed.	Good for wide-area context, too coarse for road-level segmentation alone.

Dataset	Status	How to actually get it	Caveat
Resourcesat-2 LISS-IV (5.8m)	■ Free, but gated	ISRO Bhoonidhi portal (bhoonidhi.nrsc.gov.in). Register now — under Indian Space Policy 2023 many products are instant 'Open Data' downloads, others go through a delayed-order flow.	Registration approval isn't always instant — don't leave this to the last day.
Cartosat-3 (high-res)	■ Not available pre-event	Listed on Bhoonidhi only as a priced/restricted product. The problem statement explicitly states it is provided ONLY during the 30-hour hackathon window.	Plan the whole prototype assuming you will NOT touch real Cartosat-3 data until event day.

3. The Cartosat-3 Gap — What We Actually Do About It

- **Train & validate on SpaceNet + DeepGlobe** — both are sub-meter resolution, the closest legally open proxy to Cartosat-3's resolution class.
- **Use real Bengaluru OSM road vectors + Sentinel-2/LISS-IV imagery** for the India-context demo and for Phase II/III (graph healing, centrality) — these phases don't need sub-meter pixels to work correctly.
- **Build a synthetic occlusion injector** (Albumentations + a custom cloud/shadow/tree-canopy mask generator) and overlay it on SpaceNet/DeepGlobe tiles. This lets you rigorously demonstrate and quantify 'occlusion robustness' without ever needing Cartosat data before the event.
- **Keep the data-loader decoupled from the model/graph code.** When the hackathon hands you real Cartosat-3 tiles on event day, swapping the input source is a config change, not a redesign.
- **Register on Bhoonidhi today** (free) so LISS-IV access isn't a last-minute blocker during the live event.

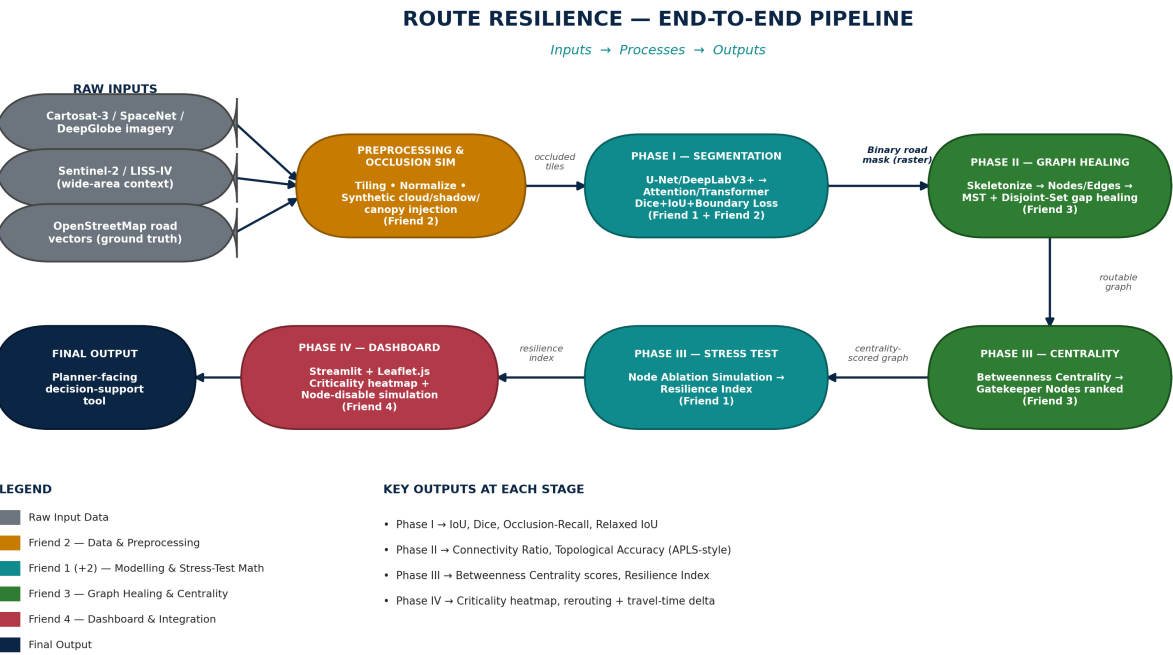
4. System Architecture — Who Touches What

Work is split so every person owns a vertical slice across phases, not one isolated block. This keeps the load equal and means no single person is siloed into just-modelling or just-frontend — everyone has both a build task and a math/logic task.

Phase	What it does	Owner(s)
I — Occlusion-Robust Segmentation	Satellite tile in → clean binary road mask out, robust to canopy/shadow/clutter.	Friend 1 + Friend 2
II — Graph Skeletonization & Healing	Mask → 1-pixel centerlines → nodes/edges → MST + Disjoint-Set gap healing into a routable graph.	Friend 3
III — Centrality & Stress Testing	Betweenness centrality → Gatekeeper Nodes → node ablation → Resilience Index.	Friend 3 + Friend 1
IV — Interactive Dashboard	Streamlit + Leaflet.js: criticality heatmap + click-to-disable-node simulation.	Friend 4
Data, Augmentation & Evaluation Backbone	Dataset acquisition, tiling, occlusion simulation, all metrics (IoU/Dice/Connectivity Ratio/Topological Accuracy).	Friend 2
Integration & Storytelling	End-to-end pipeline wiring, demo video, Idea PPT.	Friend 4 (all contribute slides)

5. End-to-End Pipeline — Inputs, Processes & Outputs (Visual)

One-glance view of how raw satellite/vector inputs flow through each phase into the final planner-facing tool, with the metric/output produced at every handoff and who owns each block.



6. Work Split — 4 People, Equal Load, Cross-Phase Ownership

Each person gets one primary build task and one analytical/logic task, so the work is balanced in both volume and difficulty — nobody is stuck only training models or only doing frontend.

FRIEND 1 (YOU) — Segmentation Modelling + Resilience & Stress-Testing Math

Tools/Libraries: PyTorch, Albumentations, NetworkX, U-Net/ResNet → UNet++/DeepLabV3+ → attention/Transformer

- Build the baseline U-Net/DeepLabV3+ on SpaceNet+DeepGlobe; establish IoU/Dice baseline and document failure cases on occluded tiles.
- Upgrade to a context-aware architecture (attention blocks / lightweight Transformer encoder) for long-range context under canopy/shadow, with combined Dice + IoU + boundary-aware (+ optional connectivity) loss.
- Once Friend 3 hands you the healed graph, implement the Node Ablation Simulation: iteratively remove top-centrality nodes and recompute global network efficiency.
- Calculate the Resilience Index (baseline avg. shortest path ÷ perturbed avg. shortest path) and package it as the 'what-if disaster' output that feeds the dashboard.
- Export masks in the exact raster format Friend 3 needs, and export ablation results in the format Friend 4 needs for the simulation toggle.

FRIEND 2 — Data Pipeline, Co-Modelling & Evaluation

Tools/Libraries: Rasterio, GDAL, OpenCV, NumPy, Albumentations, pandas, PyTorch

- Own dataset acquisition end-to-end: SpaceNet (AWS CLI), DeepGlobe (Kaggle), OSM Bengaluru pull (OSMnx), Bhoonidhi registration + LISS-IV requests.
- Build the tiling/normalization pipeline and the synthetic occlusion-injection module (simulate clouds/shadows/canopy/vehicles at controllable severity), and the train/val/test split for the whole team.
- Pair with Friend 1 on the segmentation model: co-own multi-scale feature fusion and the occlusion-handling strategy (optionally a lightweight inpainting head for heavily occluded patches).
- Own all evaluation metrics: IoU, Dice, Occlusion-Recall, Connectivity Ratio (largest connected component before/after MST healing), Topological Accuracy (APLS-style shortest-path error vs OSM), Relaxed/Length-Complete IoU with a 3–5px buffer.
- Produce the final metrics table/plots that go straight into the idea PPT.

FRIEND 3 — Graph Healing + Centrality Analysis

Tools/Libraries: scikit-image (skeletonize), NetworkX, Shapely, OSMnx

- Skeletonize Friend 1's road masks into 1-pixel centerlines; generate nodes at intersections/endpoints, edges as weighted segments.
- Implement the topological healing step: Minimum Spanning Tree + Disjoint Set to bridge gaps, scored on Euclidean distance + angular alignment so healed roads follow a natural trajectory.
- Compute Betweenness Centrality across the graph to flag 'Gatekeeper Nodes' / bottlenecks, and rank nodes for the ablation step.
- Hand Friend 1 the healed graph with centrality scores attached (for the stress-testing step), and hand Friend 4 the same graph in GeoJSON/NetworkX format for the dashboard.
- Validate the healed graph against OSM Bengaluru topology as a sanity check before handing it downstream.

FRIEND 4 — Dashboard, Integration & Storytelling Lead

Tools/Libraries: Streamlit, Leaflet.js, Folium/Plotly, PowerPoint

- Scaffold the Streamlit + Leaflet.js dashboard early (Day 1-2) using dummy/mock graph data so integration isn't blocked on others finishing first.
- Build the criticality heatmap overlay (color-coded road segments by 'Criticality Worth') from Friend 3's centrality output.
- Build the Simulation Toggle: click a node → instantly show rerouted paths + estimated travel-time increase, using Friend 1's ablation/Resilience Index output.
- Own final end-to-end integration: wire Friend 1's masks → Friend 3's graph → Friend 1's stress-test results → live dashboard, and own the demo video recording.
- Lead the Idea PPT — structure, visuals, and narrative (problem → approach → architecture diagram → results → impact); collect a results slide from each teammate.

7. Day-by-Day Roadmap (24 June → 1 July)

This is the build calendar for the prototype + PPT submission deadline. The separate 30-hour live-event run-sheet (Section 7) is what you'll follow on the actual hackathon day, once real Cartosat-3 data is handed out.

Day 1 — 24/25 Jun — Setup

- All: repo scaffold, shared Drive/Notion, finalize tech stack.
- F2: download SpaceNet + DeepGlobe; register on Bhoonidhi; pull Bengaluru OSM graph.
- F1: set up training env (Colab/local GPU), get baseline U-Net training on a small subset.
- F3: prototype skeletonization on a sample OSM/SpaceNet mask using scikit-image.
- F4: scaffold Streamlit app + Leaflet map shell with dummy markers.

Day 2-3 — 26/27 Jun — Core build

- F1 + F2: finish baseline U-Net/DeepLabV3+ together, then push into attention/Transformer upgrade + loss engineering + occlusion-handling strategy.
- F2: in parallel, finish the occlusion-injection module + augmentation pipeline and the train/val/test split.
- F3: implement MST + Disjoint Set healing; get Betweenness Centrality running on the Bengaluru OSM subgraph (doesn't need F1's output yet).
- F4: wire dashboard to F3's mock graph; build heatmap rendering logic.

Day 4 — 28 Jun — Integration Checkpoint 1

- F1 hands real predicted masks to F3.
- F3 runs skeletonization + healing on real model output for a Bengaluru AOI, computes centrality, and hands the scored graph back to F1.
- F2 runs first full metrics pass (IoU/Dice/Connectivity Ratio) and reports gaps to F1.

Day 5 — 29 Jun — Wire it live

- F1: runs the Node Ablation Simulation on F3's scored graph and calculates the Resilience Index; hands results to F4.
- F4: swaps dummy graph for F3's real Bengaluru graph + F1's stress-test output in the dashboard; builds the node-disable simulation.
- F2: finalizes Topological Accuracy (APLS-style) + Relaxed IoU numbers for the PPT.

Day 6 — 30 Jun — End-to-end run + PPT

- All: run the full pipeline on a final Bengaluru AOI; fix integration bugs.
- F4: record demo video/GIFs; assemble Idea PPT with one results slide per teammate.
- All: rehearse a 3–5 min walkthrough.

Day 7 — 1 Jul — Submit

- Final polish, package repo + PPT + demo video/screenshots.
- Submit early in the day — don't wait for the deadline hour.

8. Live Event — 30-Hour Run Sheet (once Cartosat-3 is handed out)

Hours	Focus	Who
0–2	Swap data-loader to real Cartosat-3 tiles for the assigned AOI; sanity-check resolution/CRS.	F1 + F2
2–10	Fine-tune the pre-trained segmentation model on real Cartosat-3 imagery; re-run occlusion tests.	F1 + F2
2–10 (parallel)	Run graph healing + centrality on real-imagery-derived masks as they arrive; keep dashboard live on latest output.	F3 + F4
10–20	Full pipeline integration on the event's target city/AOI; F1 runs stress-testing/Resilience Index on the live graph.	All
20–26	Polish dashboard, fix bugs, finalize metrics, build/refresh the pitch deck with live results.	All
26–30	Rehearse demo, prepare backup screenshots/video in case of live-demo failure, submit.	All

9. Tech Stack Recap

- **Preprocessing:** Rasterio, GDAL, OpenCV, NumPy, Albumentations
- **Segmentation:** PyTorch, U-Net + ResNet backbone → UNet++ / DeepLabV3+ → attention/Transformer upgrade
- **Skeletonization:** scikit-image, OSMnx
- **Graph logic:** NetworkX (core), PyTorch Geometric (optional GNN stretch goal)
- **Centrality:** NetworkX betweenness centrality
- **Visualization:** Streamlit, Leaflet.js, Matplotlib/Plotly

10. Risk Register

Risk	Mitigation
Cartosat-3 unavailable pre-event	Train/demo on SpaceNet + DeepGlobe; synthetic occlusion injector proves the concept independent of the data source (Section 3).
GPU compute limits for Transformer training	Start with a lighter attention-augmented U-Net; reserve full Transformer architecture as a stretch goal if time/compute allows.
Integration bugs between Phase I → II → III → IV	Two hard integration checkpoints (Day 4 and Day 5) instead of a single big-bang merge at the end.
Bhoonidhi LISS-IV approval delay	Register on Day 1, not later; have OSM + Sentinel-2 as a fallback that needs no approval.
Live-demo failure during judging	Keep a recorded video + static screenshots of every dashboard state as backup.